Name: _____

TOPIC TEST 2

Chapter 10: Properties of derivatives

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 6 multiple-choice questions (6 marks)
- Part B: 9 short answer questions (31 marks)
- Part C: 5 analysis questions (23 marks)

Part A

6 multiple-choice questions
1 mark each: 6 marks
Circle the correct answer.

1 When a function, $f(x)$, is decreasing, which of the following is true?

A the graph of $f(x)$ is flat

B the graph of $f(x)$ slopes upwards

C $f(x) < 0$

D the graph of $f(x)$ slopes downwards

E $f(x) = 0$

[1 mark]

2 $y = -3x^3 + 5x^2 + 16x + 20$. At $x = 2$, y is:

A increasing

B stationary

C negative

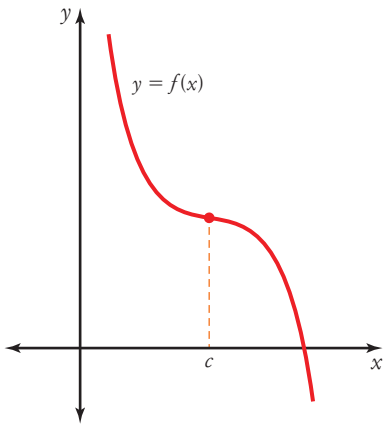
D decreasing

E positive

[1 mark]

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3 For the function shown on the right, which of the following is true?



A $f'(x) > 0$ when $x > c$

B $f'(x) < 0$

C $f'(x) < 0$ when $x < c$ and $f'(x) > 0$ when $x > c$

D $f'(x) < 0$ when $x \neq c$

E $f'(x) > 0$

[1 mark]

4 If $f(x) = 7x^2 - 8x + 5$, then $f'(2) =$

A 36

B 17

C 20

D 49

E 3

[1 mark]

5 The rate of change of $f(x) = -4x^3 + 10x^2 - 15$ at $x = -2$ is:

A -88

B -8

C 57

D 88

E 8

[1 mark]

6 The antiderivative of $-24x^5$ is:

A $120x^4 + c$

B $-4x^6$

C $6x^6 + c$

D $-4x^6 + c$

E $-6x^6 + c$

[1 mark]

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Part B

9 short answer questions
31 marks
Show your working where appropriate.

1 State whether the function $f(x) = -3x^4 + \frac{11x^3}{3} + \frac{15x^2}{2} - 17$ is increasing, decreasing or stationary at:

a $x = 2$ _____

b $x = -\frac{3}{4}$ _____

c $x = 1$ _____

[7 marks]

2 $p(x) = -5x^4 + 3x^3 + 7x - 16$. Find:

a $p'(x)$ _____

b $p'(2)$ _____

[2 marks]

3 If $f'(x) = 2x^2 + 9x - 14$ and $g'(x) = 4x^3 - 5x^2 + x$, calculate the derivative of $5f(x) - 2g(x)$.

[4 marks]

4 Find the derivatives of the following functions.

a $g(x) = 8x^4 - 12x^3 + 7x + 14$

b $h(m) = 14m - 8m^3 - 12 + 7m^2$

[2 marks]

5 Find the derivative of the function $y = (5x^2 + 3x)^2$.

[2 marks]

6 Find the derivative of the function $y = (x + 7)(2x - 3)^2$.

[2 marks]

7 $f(x) = 8 + 10x - 4x^2 + 5x^3$.

a Find the average rate of change of $f(x)$ from $x = 0$ to $x = 4$.

b Find the instantaneous rate of change at:

i $x = 0$ _____

ii $x = 4$ _____

iii $x = 2$ _____

c Compare the average rate of change and the instantaneous rates of change.

[8 marks]

8 Find two antiderivatives of $f(x) = 10x^4 + 6x^2 - 12$.

[2 marks]

9 Find y if $\frac{dx}{dy} = 21x^6 - 9x^2 + 4x$.

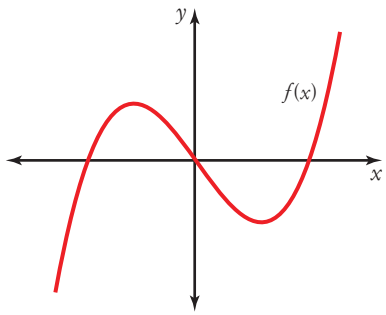
[2 marks]

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Part C

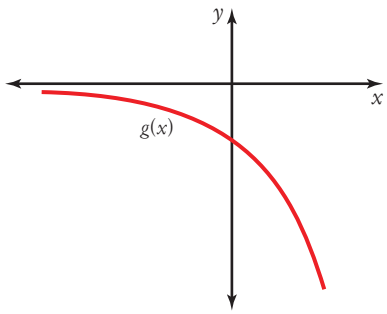
5 analysis questions
23 marks
Show your working where appropriate.

1 The graph of a function is shown on the right.
Use it to sketch a graph of the corresponding gradient function.



[4 marks]

2 The graph of a gradient function is shown on the right.
Use it to sketch a graph of the corresponding function.



[4 marks]

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3 Find the equation of the tangent to the curve $y = x^4 - 3x^3 + 7x - 2$ at $x = 2$.

[5 marks]

4 A subatomic particle passes through a linear accelerator. At speeds of less than 3×10^7 m/s its distance in metres from the start is given by:

$s = 5 \times 10^{10} t^2 + 10\,000t$, where t is in seconds.

The linear accelerator is about 8 km long.

The derivative of distance is the rate of change of distance and so it is a measure of the particle's speed.

Find the time taken for the particle to reach a speed of 3×10^7 m/s and the distance that the particle must travel through the accelerator to reach that speed.

[5 marks]

5 The rate of water flow into a dam is given by $R = 500 + 20t$ Lh^{-1}. If there is 15 000 L initially in the dam, how much water will the dam hold after 10 hours?

[5 marks]
Total marks: 60

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Answers

Part A [1 mark each]

1 The derivative is negative so the graph slopes downwards.

$\therefore$ **E** **[1 mark]**

2 $y' = -9x^2 + 10x + 16$, so when $x = 2$, $y' = 0$

$\therefore$ y is stationary.

$\therefore$ **B** **[1 mark]**

3 The derivative is the same as the gradient (slope). The function has a negative gradient for all values of x except when $x = c$, where the graph is stationary.

$\therefore$ **D** **[1 mark]**

4 $f'(x) = 14x - 8$ so $f'(2) = 20$.

$\therefore$ **C** **[1 mark]**

5 The derivative gives the rate of change. $f'(x) = -12x^2 + 20x$ so $f'(-2) = -88$.

$\therefore$ **A** **[1 mark]**

6 $f(x) = -24x^5$, so $F(x) = -\dfrac{24x^{5+1}}{6} + c = -4x^6 + c$.

$\therefore$ **D** **[1 mark]**

Part B

1 $f(x) = -3x^4 + \dfrac{11x^3}{3} + \dfrac{15x^2}{2} - 17$

$f'(x) = -12x^3 + 11x^2 + 15x$

$= x(-12x^2 + 11x + 15)$ **[1 mark]**

a $f'(2) = 2(-12 \times 2^2 + 11 \times 2 + 15) = -22$ **[1 mark]**

$f'(2) < 0$

$\therefore$ $f(x)$ is decreasing. **[1 mark]**

b $f'\left(-\dfrac{3}{4}\right) = -\dfrac{3}{4}\left[-12 \times \left(\dfrac{3}{4}\right)^2 + 11 \times \left(-\dfrac{3}{4}\right) + 15\right] = 0$ **[1 mark]**

$f'\left(-\dfrac{3}{4}\right) = 0$

$\therefore$ $f(x)$ is stationary. **[1 mark]**

c $f'(1) = 1(-12) \times 1^2 + 11 \times 1 + 15 = 14$ **[1 mark]**

$f'(1) > 0$

$\therefore$ $f(x)$ is increasing. **[1 mark]**

[7 marks]

2 **a** $p'(x) = -20x^3 + 9x^2 + 7$ **[1 mark]**

b $p'(2) = -20 \times (2)^3 + 9 \times 2^2 + 7 = -117$ **[1 mark]**

[2 marks]

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3 $D[5f(x) - 2g(x)] = 5f'(x) - 2g'(x)$ **[1 mark]**

$= 5(2x^2 + 9x - 14) - 2(4x^3 - 5x^2 + x)$ **[1 mark]**

$= 10x^2 + 45x - 70 - 8x^3 + 10x^2 - 2x$ **[1 mark]**

$= -8x^3 + 20x^2 + 43x - 70$ **[1 mark]**

[4 marks]

4 a $g'(x) = 32x^3 - 36x^2 + 7$ **[1 mark]**

b $h'(m) = 14 - 24m^2 + 14m$ **[1 mark]**

[2 marks]

5 $y = (5x^2 + 3x)^2 = 25x^4 + 30x^3 + 9x^2$ **[1 mark]**

$y' = 100x^3 + 90x^2 + 18x$ **[1 mark]**

[2 marks]

6 $y = (x + 7)(2x - 3)^2$

$= (x + 7)(4x^2 - 12x + 9)$

$= 4x^3 + 16x^2 - 75x + 63$ **[1 mark]**

$y' = 12x^2 + 32x - 75$ **[1 mark]**

[2 marks]

7 a $f(0) = 8 + 10 \times 0 - 4 \times 0^2 + 5 \times 0^3 = 8$ **[1 mark]**

$f(4) = 8 + 10 \times 4 - 4 \times 4^2 + 5 \times 4^3 = 304$ **[1 mark]**

Average rate of change $= \dfrac{f(4) - f(0)}{4 - 0} = \dfrac{304 - 8}{4} = 74$ **[1 mark]**

b Instantaneous rate of change $= f'(x) = 10 - 8x + 15x^2$ **[1 mark]**

i $f'(0) = 10 - 0 + 0 = 10$ **[1 mark]**

ii $f'(4) = 10 - 8 \times 4 + 15 \times 4^2 = 218$ **[1 mark]**

iii $f'(2) = 10 - 8 \times 2 + 15 \times 2^2 = 54$ **[1 mark]**

c The average rate of change is not the same as the average of the instantaneous rates. **[1 mark]**

[8 marks]

8 $F(x) = \dfrac{10x^{4+1}}{5} - \dfrac{6x^{2+1}}{3} - 12x + c = 2x^5 + 2x^3 - 12x + c$

Two possible antiderivatives are $2x^5 - 3x^3 + 2x^2 + 10$ and $2x^5 - 3x^3 + 2x^2 - 10$. **[1 mark each]**

[2 marks]

9 $y = \dfrac{21x^{6+1}}{7} - \dfrac{9x^{2+1}}{3} + \dfrac{4x^{1+1}}{2} + c = 3x^7 - 3x^3 + 2x^2 + c$ **[1 mark + 1 mark for + c]**

[2 marks]

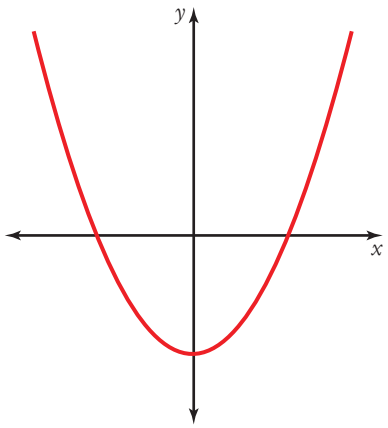
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Part C

1 The key features of the graph of the gradient function, $g(x)$, are found by examining the graph of the function, $f(x)$. These are:

- $f(x)$ is stationary on either side of the y-axis. This means that $g(x)$ will be zero at these x values.
[1 mark for showing this feature on the graph]
- $f(x)$ has a negative slope between the stationary points. This means that $g(x)$ will be negative between these x values.
[1 mark for showing this feature on the graph]
- Looking from left to right, $f(x)$ has a positive slope on the left of the first stationary point and on the right of the second stationary point. This means that $g(x)$ will be positive at these x values.
[1 mark for showing this feature on the graph]
- The steepness of the graph of $g(x)$ is determined by the steepness of the tangent to the graph of $f(x)$.
[1 mark for showing this feature on the graph]

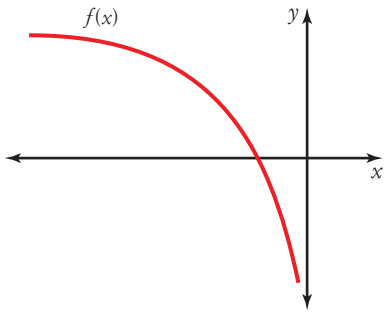
[4 marks]



2 The key features of the graph of the function, $f(x)$, are found by examining the graph of the gradient function, $g(x)$. These are:

- $g(x)$ does not intercept the x-axis. This means that $f(x)$ does not have any stationary points.
[1 mark for showing this feature on the graph]
- $g(x)$ is negative for all values of x. This means that $f(x)$ is decreasing for all values of x.
[1 mark for showing this feature on the graph]
- The values of $g(x)$ indicate the steepness of the tangent to the graph of $f(x)$. Higher values of $g(x)$ indicate that the graph of $f(x)$ will be steeper than at lower values of $g(x)$.
[2 marks for showing this feature on the graph]

[4 marks]



3 When $x = 2$, $y = 2^4 - 3 \times 2^3 + 7 \times 2 - 2 = 4$ **[1 mark]**

$y' = 4x^3 - 9x^2 + 7$

When $x = 2$, $y' = 4 \times 2^3 - 9 \times 2^2 + 7 = 3$ **[1 mark]**

The tangent passes through (2, 4) and has a gradient of 3.

Equation of the tangent: $y - y_1 = m(x - x_1)$ **[1 mark]**

$(y - 4) = 3(x - 2)$ **[1 mark]**

$y - 4 = 3x - 6$

$y = 3x - 2$ **[1 mark]**

[5 marks]

4 $v = \dfrac{ds}{dt} = 2 \times 5 \times 10^{10}t + 10\,000 = 10^{11}t + 10\,000$ **[1 mark]**

Let $v = 3 \times 10^7$

$3 \times 10^7 = 10^{11}t + 10\,000$ **[1 mark]**

$10^{11}t = 3 \times 10^7 - 10\,000$

$10^{11}t = 2.999 \times 10^7$

$t \approx 2.999 \times 10^{-4}$ **[1 mark]**

Let $t \approx 2.999 \times 10^{-4}$:

$s = 5 \times 10^{10}\,(2.999 \times 10^{-4})^2 + 10\,000 \times 2.999 \times 10^{-4}$

≈ 4500 **[1 mark]**

It takes about 0.3 milliseconds and about 4.5 km for the particle to reach a speed of 3×10^7 m/s. **[1 mark]**

[5 marks]

5 The antiderivative of R gives an expression for the volume of water in the dam. **[1 mark]**

Let V be the antiderivative of R.

$V = 500t + 10t^2 + c$ **[1 mark]**

$V = 15\,000$ at $t = 0$, so $15\,000 = 500 \times 0 + 10 \times 0^2 + c$

$c = 15\,000$ so $V = 500t + 10t^2 + 15\,000$ **[1 mark]**

Let $t = 10$: $V = 500 \times 10 + 10 \times 10^2 + 15\,000 = 21\,000$ **[1 mark]**

The volume of the dam after 10 hours is 21 000 L. **[1 mark]**

[5 marks]

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